

Normal and Hermitian.

Lemma. Let  $V$  be an inn. prod space and  $T$  is a linear operator.

(If exists) Every eigenvalue of  $T$  is real.

Moreover, eigenvectors of  $T$  corresponding to distinct eigenvalues are orthogonal.

Proof. Given eigenvalue  $C$  of  $T$  corresponding to  $\alpha$ ,

$$C \cdot \langle \alpha, \alpha \rangle = \langle T(\alpha), \alpha \rangle = \langle \alpha, T^*(\alpha) \rangle = \langle \alpha, T(\alpha) \rangle = \langle \alpha, C \cdot \alpha \rangle = \bar{C} \cdot \langle \alpha, \alpha \rangle$$

Since  $\alpha$  is non-zero,  $C = \bar{C}$ , or  $C$  is a real number.

And, take  $\alpha_1, \alpha_2$  be eigenvectors corresponding to  $C_1, C_2$ , with  $C_1 \neq C_2$ .

Then,

$$C_1 \cdot \langle \alpha_1, \alpha_2 \rangle = \langle C_1 \alpha_1, \alpha_2 \rangle = \langle T(\alpha_1), \alpha_2 \rangle = \langle \alpha_1, T^*(\alpha_2) \rangle = \langle \alpha_1, T(\alpha_2) \rangle = \langle \alpha_1, C_2 \alpha_2 \rangle = \bar{C}_2 \cdot \langle \alpha_1, \alpha_2 \rangle$$

Since  $C_2$  must be real,  $\bar{C}_2 = C_2$ . Now, since  $C_1 \neq C_2$ ,

$$(C_1 - C_2) \cdot \langle \alpha_1, \alpha_2 \rangle = 0 \text{ implies } \langle \alpha_1, \alpha_2 \rangle = 0.$$

Thm. In finite-dimensional inn. prod space,

Every Hermitian operator has an eigenvector.

Proof. If the given field is  $\mathbb{C}$ , then the characteristic polynomial must have a root in  $\mathbb{C}$  by the Fundamental Theorem of Algebra. So it has an eigenvector.

Now, assume that the given field is the  $\mathbb{R}$ .

Let  $T$  be a Hermitian on  $V$  over  $\mathbb{R}$ , and  $\beta$  be an orthonormal basis for  $V$ .

Let  $A = [T]_{\beta}$ . Since  $T = T^* = T^t$ ,  $A = A^* = A^t$ . Define an inner product on  $\mathbb{C}^n$  as:

$$\langle X, Y \rangle \stackrel{\text{def}}{=} Y^* X$$

Then, a linear operator  $L_A: \mathbb{C}^n \rightarrow \mathbb{C}^n: X \mapsto AX$  is Hermitian, because

$$\langle L_A(X), Y \rangle = \langle X, L_A^*(Y) \rangle \text{ and } \langle AX, Y \rangle = Y^* AX = \langle X, A^* Y \rangle = \langle X, AY \rangle.$$

By the FTA,  $L_A$  has an eigenvalue, and that must be real, by the above Lemma.

Now,  $L_A(X) = AX = CX$  for some real number  $C$  and a non-zero vector  $X$  in  $\mathbb{C}^n$ .

or  $\det(CI - A) = 0$ ,  $A - CI$  is not invertible, and  $A - CI$  has only real-entries,

so  $(A - CI)X = 0$  has a  $\underset{\text{non-zero}}{\text{real solution}} X$ .

Lemma. Let  $V$  be a fin. dim inn. prod space, and  $T$  be a linear operator on  $V$ .

If  $W \subset V$  is  $T$ -invariant, then  $W^\perp$  is  $T^*$ -invariant.

Proof. Given  $\beta \in W^\perp$ , and  $\alpha \in W$ ,  $\langle T^*(\beta), \alpha \rangle = \langle \beta, T(\alpha) \rangle = 0$ , by  $T(\alpha) \in W$ .  $\square$

Diagonalizable Criterion of linear operator on inner product space.

Thm. Let  $T$  be a linear operator on fin. dim inner prod space  $V$ .

If  $T$  is Hermitian, then there is an orthonormal basis for  $V$  consisting by eigenvectors of  $T$ .

Proof. Let  $T$  be a Hermitian. Using induction for  $\dim V = n$ .

If  $n=1$ , trivial. And, for  $n$ , using the facts that:

- 1).  $T$  has an eigenvector  $\alpha_1$ .
- 2).  $\langle \alpha_1 \rangle \oplus \langle \alpha_1 \rangle^\perp = V$ .
- 3).  $\langle \alpha_1 \rangle$  is  $T$ -invariant, so  $\langle \alpha_1 \rangle^\perp$  is  $T^*$ -invariant, and  $T = T^*$ .

Corollary. In  $\mathbb{R}$ -Inner Product space,

$T$  is symmetric if and only if there is orthonormal basis for  $V$  consisting by eigenvectors of  $T$ .

Lemma. Let  $T$  be a normal operator on fin. dim inner prod space  $V$ . Given  $\alpha \in V$ ,

$\alpha$  is eigenvector for  $T$  corresponding to  $c \in \mathbb{F}$  iff  $\alpha$  is eigenvector for  $T^*$  corresponding to  $\bar{c} \in \mathbb{F}$ .

Proof. If  $L$  is normal, then

$$\|L(\alpha)\|^2 = \langle L(\alpha), L(\alpha) \rangle = \langle \alpha, L^*L(\alpha) \rangle = \langle \alpha, LL^*(\alpha) \rangle = \langle L^*(\alpha), L^*(\alpha) \rangle = \|L^*(\alpha)\|^2.$$

And, any scalar  $c \in \mathbb{F}$ ,  $T - cI$  is normal, by

$$(T - cI)(T - cI)^* = T^*T - \bar{c}T - cT^* + c\bar{c} = T^*T - \bar{c}T - cT^* + c\bar{c} = (T - cI)^*(T - cI).$$

So,  $\|(T - cI)(\alpha)\| = \|(T - cI)^*(\alpha)\|$  gives the result.

Thm. Let  $T$  be a linear operator on fin. dim  $V$  s  $V$ , and  $\beta$  be an orthonormal basis for  $V$ .

Suppose that  $A = [T]_\beta$  is upper-triangular. Then,

$T$  is normal if and only if  $A$  is diagonal.

Proof. Let  $T$  is normal, and  $\beta = \{\alpha_1, \dots, \alpha_n\}$ . Since  $A$  is upper triangular,

$$T(\alpha_1) = A_{11}\alpha_1 \text{ and so } T^*(\alpha_1) = \bar{A}_{11}\alpha_1 = \sum_{i=1}^n (A^*)_{i1} \cdot \alpha_i = \sum \bar{A}_{i1} \cdot \alpha_i$$

So,  $\bar{A}_{12} = \bar{A}_{13} = \dots = \bar{A}_{1n} = 0$ . Similarly, all non-diagonal entries are zero.

The Converse is trivial. ~~QED~~

Now, since every linear operator on  $\mathbb{C}$ -vector space has a basis s.t. the representation is triangular. Using Gram-Schmidt process to this, we obtain that every normal in  $\mathbb{C}$  is unitary diagonalizable.

Summary.

$T: V \rightarrow V$ ,  $V$  fin. dim inn. prod over  $F$

$T$  is Normal and  $F = \mathbb{C}$

In  $\mathbb{C}$ ,  $T$  is Hermitian  $\Rightarrow T$  is unitary equivalent to a diagonal matrix  $\Downarrow$

In  $\mathbb{R}$ ,  $T$  is symmetric  $\Leftrightarrow T$  is orthogonal equivalent to a diagonal matrix

In  $\mathbb{C}$ , Hermitian  $\Rightarrow$  Normal  $\Leftarrow$   
 $\Downarrow$   
Symmetric